

CS 4412 — M2 Summary (Initial Implementation)

Project: Discovering Short-Term Fatigue-Management Patterns (Power Naps vs Coffee)

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Goal (discovery, not prediction): demonstrate deep understanding of the dataset via EDA + preprocessing, then apply **one** mining technique and explain at least one pattern discovered.

Dataset: 500 rows × 11 columns (CSV). Each row is a simulated fatigue-management session with prior sleep, intervention type/duration, alertness before/after, productivity, mood, and side effects.

Preprocessing & Feature Engineering (decisions):

- Filled missing *side_effects* with “None/Not Reported” to avoid dropping rows and to preserve missingness information.
- Removed exact duplicate rows (if any).
- Created *alertness_delta* = *alertness_after* – *alertness_before* (core discovery feature).
- For clustering, standardized numeric features (z-score) so scale does not dominate distance.

Key EDA snapshots:

Sleep hours (prev night): mean=6.059, min=4.0, max=8.0.

Intervention duration (min): mean=22.3, min=15.0, max=30.0.

Alertness delta: mean=13.85, min=5.0, max=24.0.

Intervention	Mean alertness_delta	Mean productivity	Mean mood
Coffee	10.927	5.349	5.517
Power Nap	17.042	5.402	5.318

Mining Technique (M2): K-Means Clustering

We cluster sessions using numeric features: sleep hours, duration, alertness before/after, *alertness_delta*, productivity, and mood. The notebook computes silhouette scores for $k=2..8$ and fits K-Means with the best k . Cluster profiles (means + category proportions) are reported to support interpretation.

Preliminary Findings (what to report): clusters represent different response patterns (e.g., higher delta vs lower delta groups). The M2 requirement is to describe at least one cluster pattern and explain why it matters (“So what?”).

Updated plan for M3:

- Add association rules: discretize sleep, duration, delta, productivity, mood; mine rules with support/confidence/lift; focus on trade-offs and side effects.
- Add anomaly detection: flag unusually low/negative delta or unusually high delta for similar profiles; explain why they stand out.
- Use PCA plots to visualize clusters and anomalies; compare K-Means with DBSCAN or hierarchical clustering for robustness.